

B1 — Data Source Inventory and Quality Assessment

Part A — Data Source Inventory

Data Source	Grain Statement
branches.csv	One row represents one Kape Tayo branch and its location information.
calendar.csv	One row represents one calendar date and its corresponding date attributes.
customers.csv	One row represents one registered Kape Tayo customer.
employees.xlsx	One row represents one employee and their branch, role, rate, and employment information.
products.xlsx	One row represents one product, including its category, price, and cost.
inventory_movements.csv	One row represents one inventory movement for a product at a branch on a specific date.
sales_transactions.csv	One row represents one recorded sales transaction involving a branch, product, quantity, amount, and payment method.

Part B — Six Business Questions

- 1. Which branch generated the most sales revenue since its opening date?**
- 2. Do the number of staff directly address the branch needs? If not, why?**
- 3. What underlying factor causes an increase in sales in a day?**

4. Do customer loyalty services generate more repeat customers?
5. Is the inventory for every branch aligned with its daily demand?
6. Which product sells more in a specific branch? Does

Part C — Data Quality Assessment

Dimension	Assessment
Completeness	Some information is missing. Sales transactions contain missing CustomerIDs and several missing dates, while some customers have no email and some employees have no EmploymentStatus. The calendar also has missing dates.
Uniqueness	Most master records have unique IDs, but the sales data contains 23 duplicate SaleIDs that could overstate sales if not removed.
Validity	Some values do not follow expected business rules. Examples include inconsistent BranchID formats, ProductID P99 with no matching product, inconsistent customer tiers, an employee assigned to nonexistent Branch B07 , and one negative employee daily rate.
Consistency	Some records do not match correctly across datasets. Missing calendar dates prevent some sales and inventory transactions from joining properly. Customer tier and inventory movement labels also use inconsistent formatting.
Accuracy	Some sales records have amounts that do not equal Quantity × Unit Price , including transactions with zero quantity but positive revenue. Some unusually large inventory deliveries should also be verified before analysis.

Timelines	The operational data mainly ends in June 2026 , so it is useful for historical analysis but does not fully represent Kape Tayo's current August 2026 situation.
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Part D — Verdict

The datasets provide a useful foundation for sales, customer spending behavior, product, inventory utilization, employee, and branch performance analysis. However, some of the datasets have data-quality issues that could potentially cause discrepancies in my analysis.

To achieve my desired outcome, I would need to remove duplicates, fix invalid IDs, complete missing information, and validate questionable sales. Only after cleaning the datasets will I be able to confidently present my analysis

Milestone	Date	Tool	Purpose	What it produced	How I verified or changed it	Tier
B1	24/08/2026	ChatGPT	Data content analysis.	It produced the desired outcome to identify which data are redundant and which data are missing.	When it gave me the answers, I verified them by looking at the data and based my analysis on the outcome.	

